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Course Code: MAT337

DATA SCIENCE USING PYTHON

Course Objectives:

The course aims to train the learner to:

- Understand the core principles of scientific computing and data science using Python.
- Apply Python programming constructs to solve mathematical and computational problems.
- Implement numerical and statistical algorithms using libraries such as NumPy, SciPy, and pandas.
- Analyze datasets through data manipulation, transformation, and exploratory data analysis techniques.
- Apply Python tools to work with real-world datasets and draw meaningful insights.

UNIT I: 12 Periods

Python Foundations — Python syntax and semantics, control structures, functions, and scripting. Data types (numeric, string, list, dict, etc.) and their use in algorithmic problem- solving. Basic I/O and file handling for data processing.

UNIT II 12 Periods

Data Structures & Algorithms in Python — Tuples, sets, dictionaries; list comprehensions and generator expressions. Complexity of operations on lists vs. numpy arrays. Implementing simple mathematical algorithms (e.g. finding roots, prime generation) to illustrate efficiency and correctness. Introduction to using Python notebooks for interactive computing.

UNIT III: 12 Periods

Numerical Computing with NumPy — Ndarray structure and operations: vectorization of computations, element-wise vs. matrix operations, broadcasting. Linear algebra operations (dot product, matrix decompositions) using NumPy's linalg module. Solving linear systems and basic optimization problems in Python. Precision and performance considerations (floating-point issues, complexity analysis).

UNIT IV: 12 Periods

Data Analysis with pandas — Reading/writing data (CSV, JSON), cleaning and preprocessing data. Data frames, indexing, grouping, and aggregations. Descriptive statistics and probability distributions using Python. Case studies: e.g. analyzing a public dataset (economic or health data) using pandas. Integration with NumPy for statistical computations

UNIT V: 12 Periods

Visualization and Intro to ML — Plotting with Matplotlib/Seaborn: line chans, histograms, scatter plots, heatmaps for correlation. Designing clear visualizations of data trends. Basic machine learning workflow in Python: using scikit-learn to perform a simple regression or classification (e.g. linear regression on a dataset, or k-means clustering) and evaluating the results. Introduction to Jupyter notebooks as a tool for iterative data science.

Time series basics: DateTime index, resampling, rolling/expanding statistics; simple baseline forecasts (naive/seasonal-naive) and error metrics.

List of Experiments for Practical Sessions:

1. Writing Python scripts for math puzzles
2. Implementing matrix operations from scratch vs. using NumPy
3. Exploratory data analysis on sample datasets. and a mini-project (e.g. analyzing a real dataset from Kaggle or UCI repository).
4. Emphasize problem-solving, code documentation, and correct use of libraries.

REFERENCES

1. Charles Severancei, *Python for Everybody: Exploring Data Using Python 3,* Shroff Publishers & Distributors (SPD), 1st Edition, 2016.
2. Wes McKinney, *Python for Data Analysis*, 3rd Edition, published by O'Reilly Media, 2022.
3. Jake Vander Plos,*Python Data Science Handbook*, 2nd Edition, O'Reilly Media , 2022.
4. Aileen Nielsen, *Practical Time Series Analysis: Prediction with Statistics and Machine Learning*, O'Reilly Media (2019).

ONLINE MATERIALS

https://nptel.ac.in/courses/106106212

LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to:

UNIT I	Use control structures (if-else, loops) to implement logic in programs. Write reusable code using functions, parameters, return values, and modular programming
UNIT II	Use advanced Python data structures such as tuples, sets, and dictionaries. Write efficient code using list comprehensions and generator expressions.
UNIT III	Perform vectorized and element-wise numerical operations efficiently. Use broadcasting rules to simplify mathematical expressions.
UNIT IV	Perform grouping, aggregation, and summarization of data.
UNIT V	Create visualizations such as line plots, histograms, scatter plots, and heatmaps using Matplotlib and Seaborn. Implement naïve and seasonal-naïve forecasting methods and evaluate forecasts using appropriate error metrics.

COURSE LEARNING OUTCOMES

Upon successful completion of the course, the student will be able to:

CO NO.	COURSE OUTCOME	KNOWLEDGE LEVEL
CO 1.	Work with Python data types such as numbers, strings, lists, tuples, dictionaries, and sets. Apply data types effectively in algorithmic problem-solving tasks.	K3
CO 2.	Compare the complexity of operations on Python lists and NumPy arrays. Implement simple mathematical algorithms (roots, primes, iteration-based methods).	K3
CO 3.	Perform linear algebra operations such as dot products, matrix multiplication, eigenvalues, and decompositions using numpy.linalg. Solve linear systems using NumPy functions.	K3
CO 4.	Compute descriptive statistics and basic probabilistic measures. Integrate pandas with NumPy to perform statistical and numerical operations	K3
CO 5.	Build simple ML models such as linear regression or k-means clustering and evaluate forecasts using appropriate error metrics.	K3